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CC Chemokine Ligands 25 and 28 Play Essential Roles in Intestinal Extravasation of IgA Antibody-Secreting Cells[1]

Kunio Hieshima,* Yuri Kawasaki,* Hitoshi Hanamoto,† Takashi Nakayama,* Daisuke Nagakubo,* Akihisa Kanamaru,† and Osamu Yoshie[2]*‡

CCL25 (also known as thymus-expressed chemokine) and CCL28 (also known as mucosae-associated epithelial chemokine) play important roles in mucosal immunity by recruiting IgA Ab-secreting cells (ASCs) into mucosal lamina propria. However, their exact roles in vivo still remain to be defined. In this study, we first demonstrated in mice that IgA ASCs in small intestine expressed CCR9, CCR10, and CXCR4 on the cell surface and migrated to their respective ligands CCL25, CCL28, and CXCL12 (also known as stromal cell-derived factor 1), whereas IgA ASCs in colon mainly expressed CCR10 and CXCR4 and migrated to CCL28 and CXCL12. Reciprocally, the epithelial cells of small intestine were immunologically positive for CCL25 and CCL28, whereas those of colon were positive for CCL28 and CXCL12. Furthermore, the venular endothelial cells in small intestine were positive for CCL25 and CCL28, whereas those in colon were positive for CCL28, suggesting their direct roles in extravasation of IgA ASCs. Consistently, in mice orally immunized with cholera toxin (CT), anti-CCL25 suppressed homing of CT-specific IgA ASCs into small intestine, whereas anti-CCL28 suppressed homing of CT-specific IgA ASCs into both small intestine and colon. Reciprocally, CT-specific ASCs and IgA titers in the blood were increased in mice treated with anti-CCL25 or anti-CCL28. Anti-CXCL12 had no such effects. Finally, both CCL25 and CCL28 were capable of enhancing α_4 integrin-dependent adhesion of IgA ASCs to mucosal addressin cell adhesion molecule-1 and VCAM-1. Collectively, CCL25 and CCL28 play essential roles in intestinal homing of IgA ASCs primarily by mediating their extravasation into intestinal lamina propria. ***The Journal of Immunology,*** **2004, 173: 3668–3675.**

Chemokines are a large group of closely related cytokines and transmembrane proteins that play important roles in innate and acquired immunity primarily by inducing directed migration of various types of leukocytes through interactions with a group of seven-transmembrane, G protein-coupled receptors (1). It is now known that chemokines and their receptors are crucial elements in the trafficking and tissue microenvironmental localization of various lymphocyte classes and subsets (1–3). Thus, in accordance with differentiation, activation, and functional maturation, lymphocytes dynamically change their expression profiles of chemokine receptors and shift their migration programs to a new set of chemokines (1–3).

Plasma cells represent the end stage of B cell differentiation and function as the factories of Ab production. Upon antigenic stimulation, naive B cells either migrate to the medullary cords, where they proliferate and rapidly differentiate into short-lived plasma cells producing low-affinity IgM Abs, or migrate to the B cell follicles, where they participate in the germinal center reaction, which leads to differentiation into long-lived plasma cells producing high-affinity IgG and IgA Abs (4). Thus, there are roughly two types of plasma cells, the short-lived ones mostly dying in situ and the long-lived ones homing to a wide variety of organs such as bone marrow and gastrointestinal tract. Furthermore, plasma cells producing IgG Abs tend to home to bone marrow, whereas those producing IgA Abs migrate into lamina propria of gastrointestinal, respiratory, and urogenital tissues (5, 6). Chemokines are likely to play important roles in the migration and tissue localization of plasma cells. In this context, Hargreaves et al. (7) reported that CXCR4 and its ligand CXCL12 were critically involved in the localization of plasma cells within the splenic red pulp and lymph node (LN)[3] medullary cords as well as in their homing to the bone marrow. Wehrli et al. (8) demonstrated down-regulation of chemokine receptors such as CCR7 and CXCR5 in plasmablasts, which presumably allows their migration from the B cell follicles to the medullary cords and into efferent lymphatic vessels. Hauser et al. (9) demonstrated that Ag-specific IgG Ab-secreting cells (ASCs), which appeared in the spleen a few days after secondary immunization and were destined to home to the bone marrow, selectively migrated to CXCR4 and CXCR3 ligands. Bowman et al. (10) reported that IgA ASCs, but not IgG or IgM ASCs, in mice expressed CCR9 and efficiently responded to its ligand CCL25, which is selectively expressed in the small intestine (11, 12). Previously, we have demonstrated that plasma cells in mouse salivary glands as well as those in human bone marrow selectively express CCR10 and respond to its ligand CCL28 (13, 14), which is selectively expressed in certain mucosal tissues and human bone marrow (13–16). Butcher and coworkers (17, 18) have also demonstrated that CCL28 attracts IgA ASCs present in both intestinal and

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Received for publication February 3, 2004. Accepted for publication July 16, 2004.

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[1] This work was supported in part by a High-Tech Research Center Grant from the Ministry of Education, Culture, Sports, Science, and Technology, Japan; by Solution Oriented Research for Science and Technology of Japan Science and Technology Agency; and by a grant from the Ministry of Health, Labor, and Welfare, Japan.

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[3] Abbreviations used in this paper: LN, lymph node; ASC, Ab-secreting cell; CT, cholera toxin; MAdCAM, mucosal addressin cell adhesion molecule; MLN, mesenteric LN; PP, Peyer's patch.

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nonintestinal mucosal tissues, while CCL25 only attracts a subpopulation of IgA ASCs associated with small intestine. These results suggest that CCL25 and CCL28 play important roles in the intestinal homing of IgA ASCs. However, their exact roles in homing of IgA ASCs in intestinal lamina propria still remain to be defined.

In this study, we have further explored the roles of CCL25 and CCL28 in intestinal homing of IgA ASCs in mice. As reported previously, CCL25 and CCL28 are strongly expressed in the small intestine and colon, respectively (11, 12, 15, 16). We have further shown that CCL28 is also expressed in mouse small intestine. Furthermore, we have revealed by immunohistochemistry that CCL25 and CCL28 are present not only in intestinal epithelial cells, but also in endothelial cells of small venules in intestinal lamina propria. This suggested direct roles of CCL25 and CCL28 in extravasation of IgA ASCs into intestinal lamina propria. Consistently, anti-CCL25 prevented homing of Ag-specific IgA ASCs into small intestine, whereas anti-CCL28 prevented their homing into both small intestine and colon. Furthermore, both CCL25 and CCL28 enhanced α_4 integrin-dependent adhesion of IgA ASCs to mucosal addressin cell adhesion molecule-1 (MAdCAM-1) and VCAM-1 in vitro. Collectively, CCL25 and CCL28 play essential roles in intestinal homing of IgA ASCs primarily by promoting their extravasation into intestinal lamina propria.

Materials and Methods

Mice

Female BALB/c mice were purchased from SLC (Hamamatsu, Japan) and maintained in specific pathogen-free conditions for at least 1 wk before experiments. All animal experiments in the present study were approved by the Center of Animal Experiments, Kinki University School of Medicine.

Chemokines and Abs

All recombinant mouse chemokines and human stromal cell-derived factor-1α/CXCL12 were purchased from R&D Systems (Minneapolis, MN). Affinity-purified goat-neutralizing Abs against mouse CCL25 (AF-481-NA), mouse CCL28 (AF533), and human CXCL12 (AF-310-NA) were also purchased from R&D Systems. Control normal goat IgG was purchased from Genzyme-Techne (Mineapolis, MN). We used goat anti-human CXCL12 for detection and neutralization of mouse CXCL12 because human and mouse CXCL12 differ only at a single amino acid residue (1). We confirmed that anti-human CXCL12 reacted equally well with both human and mouse CXCL12 in ELISA and similarly neutralized both human and mouse CXCL12 in a standard chemotaxis assay. Specifically, the neutralizing activities (ED_{50}) of anti-mouse CCL25, anti-mouse CCL28, and anti-human/mouse CXCL12 determined against mouse CCL25 at 1 μg/ml, mouse CCL28 at 2 μg/ml, and mouse CXCL12 at 2 ng/ml, respectively, in a standard chemotaxis assay were 4–12, 5–25, and 0.2–0.6 μg/ml, respectively. The concentrations of chemokines used in these assays were the optimal doses that gave maximal chemotactic responses. The relatively low neutralizing activity of anti-CXCL12 may be in part due to a very high potency of CXCL12 in comparison with CCL25 or CCL28.

Real-time PCR analysis

Real-time PCR was performed using TaqMan assay and 7700 Sequence Detection System (Applied Biosystems, Foster City, CA). Template cDNAs were generated from total RNAs extracted from various mouse tissues using guanidinium isothiocyanate (19). Amplification conditions were 50°C for 2 min, 95°C for 10 min, and 50 cycles of 95°C for 15 s (denaturation) and 60°C for 1 min (annealing/extension). The primers were as follows: +5′-AGCACAGGATCAAATGGAATGTT-3′ and −5′-GGTTGCAGCTTCCACTCACTT-3′ for CCL25; +5′-TGAGCCCGGCTCCTGAA-3′ and −5′-GCTTGGGAGTGGCTGTCTATAGA-3′ for CCL27; +5′-CAGCCCGCACAATCGTACT-3′ and −5′-ACGTTTTCTCTGCCATTCTTCTTT-3′ for CCL28; and + 5′-TGCCCCTGCCGGTTCT-3′ and −5′-TGTTGAGGATTTTCAGATGCTTGA-3′ for CXCL12. The probes for chemokines were as follows: +5′-TCCGGCATGCTAGGAATTATCACCAGC-3′ for CCL25; +5′-CTTGCCTCTGCCCTCCAGCACTAGCT-3′ for CCL27; +5′-TGAAGCAGTGGATGAGAGCCTCAGAGG-3′ for CCL28; and +5′-CGAGAGCCACATCGCCAGAGCC-3′ for CXCL12. The probes were labeled with the reporter fluorescent dye 6-FAM at the 5′ end. The primers and fluorogenic probes for 18S ribosomal RNA were obtained from TaqMan kit (Applied Biosystems). Chemokine expression was quantified by using Sequence Detector System software (Applied Biosystems).

Isolation of lymphocytes

All tissues were obtained from 8- to 12-wk-old female BALB/c mice. Lymphocytes were prepared from Peyer's patches (PPs) and mesenteric LNs (MLNs) by mincing the tissues in RPMI 1640 containing 10% FCS with scissors and pressing them through a nylon mesh screen with a round-edged spatula. Lamina propria lymphocytes were isolated from small intestine and colon, as described previously (10). Briefly, small intestines that were carefully cleared of PPs and colons were cut open longitudinally, cut into ~5-mm segments, and washed at room temperature with vigorous shaking four times in divalent cation-free HBSS supplemented with 5 mM EDTA, 25 mM HEPES, 100 U/ml penicillin, and 100 μg/ml streptomycin to remove epithelial cells and intraepithelial lymphocytes until no more shedding occurred. Intestinal tissues were washed twice in RPMI 1640 containing 10% FCS, 15 mM HEPES, and antibiotics. Then lamina propria lymphocytes were released by shaking intestinal tissues in RPMI 1640 containing 20% FCS, 25 mM HEPES, antibiotics, and 300 U/ml collagenase type VIII (Sigma-Aldrich, St. Louis, MO) for three 40-min sessions. At the end of each 40-min incubation, released cells were immediately washed in RPMI 1640 containing 10% FCS and antibiotics. Isolated lymphocytes were resuspended in RPMI 1640 containing 10% FCS and antibiotics and incubated in 6-cm dishes at 37°C in a CO_2 incubator for 2 h to remove adherent cells and to allow recovery from potential desensitization of chemokine responses.

Flow cytometric analysis

Single cells were isolated from various mouse tissues, as described above. Blood leukocytes were prepared from heparinized blood samples through osmotic lysis of RBC. Cells were suspended in ice-cold PBS containing 3% FCS and 0.1% sodium azide (staining medium). All of the following steps were conducted on ice. Cells were first treated with PBS containing 0.1% BSA, 2% normal mouse serum, and 1 μg/ml anti-mouse CD32/16 (Beckman Coulter, Marseille, France) to block Fc receptors. After washing, cells were incubated with a mixture of FITC-labeled anti-IgA (C10-3) or FITC-labeled isotype-matched control IgG, PE-labeled anti-CD3ϵ, CyChrome-labeled anti-B220, and biotinylated anti-CD38 (all from BD Pharmingen, Mountain View, CA) for 30 min. After washing, cells were incubated with streptavidin-allophycocyanin for 30 min. For surface staining of CCR9, CCR10, and CXCR4, mouse CCL25-Fc, human CCL27-Fc, mouse CXCL12-Fc, respectively, and control Fc were used, as described previously (13). After staining, cells were immediately analyzed on a FACSCalibur (BD Biosciences, Mountain View, CA).

Chemotaxis assay

This was performed using 8-μm Transwell plates (Corning Costar, Cambridge, MA), as described previously (13). Briefly, cells were placed in upper wells (2×10^6/well), while bottom wells contained medium alone or medium containing chemokines at concentrations that were predetermined to be optimal (300 nM for CCL25, 300 nM for CCL28, and 50 nM for CXCL12) (13, 14). After incubation at 37°C for 4 h, migrated cells in multiple replicate wells (generally 4–5 wells per each condition) were combined. IgA ASCs in original and migrated cell populations were enumerated by ELISPOT.

ELISPOT

Total IgA ASCs and cholera toxin (CT)-specific IgA ASCs were enumerated with ELISPOT following the protocol provided by BD Pharmingen. Briefly, nitrocellulose 96-well plates (Multiscreen 96-Well Filtration Plate; Millipore, Billerica, MA) were coated with 2 μg/ml polyclonal goat anti-mouse IgA (Kirkegaard & Perry Laboratories, Gaithersburg, MD) or CT (Sigma-Aldrich) in PBS at 4°C overnight. After washing and blocking with RPMI 1640 containing 10% FCS, cells suspended in RPMI 1640 containing 10% FCS, 2 mM L-glutamine, 100 U/ml penicillin, and 100 μg/ml streptomycin were added to wells (1×10^3 to 5×10^5 cells/100 μl/well) and incubated at 37°C overnight in humidified air with 5% CO_2. After washing with deionized water, each well was added with 100 μl of biotinylated polyclonal anti-mouse IgA (Kirkegaard & Perry Laboratories) at 100 ng/ml in PBS containing 10% FCS and incubated at room temperature for 2 h. After washing with PBS containing 10% FCS, each well was added with 100 μl of streptavidin-HRP in PBS containing 10% FCS and incubated at room temperature for 1 h. After washing thoroughly with PBS

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containing 0.05% Tween 20, each well was added with 100 μl of chromogen substrate containing 0.3 mg/ml 3-amino-9-ethylcarbazole (Sigma-Aldrich) and 0.015% (v/v) H_2O_2 and incubated for 15–20 min, yielding reddish spots by ASCs. Numbers of spots per well were counted under an inverted microscope.

Immunohistochemistry

Frozen sections were made from small intestine and colon derived from 8-wk-old female BALB/c mice and briefly fixed with periodate-lysine-4% paraformaldehyde. After washing, sections were treated with anti-CD32/16 (Beckman Coulter) supplemented with 10–50% normal rabbit serum, 3% BSA, 0.25% gelatin, 0.025% Nonidet P-40, and 5 mM EDTA, and further treated with Biotin Blocking System (DakoCytomation, Kyoto, Japan). Then sections were reacted with goat anti-mouse CCL25, goat anti-mouse CCL28, goat anti-human/mouse CXCL12, or normal goat IgG. After washing, sections were successively reacted with biotinylated rabbit anti-goat IgG (Vector Laboratories, Burlingame, CA) and Vectastain ABC/HRP kit (Vector Laboratories). Chromogenic reactions were performed with diaminobenzidine and H_2O_2, resulting in dark brown reaction products in positive cells. Sections were counterstained with hematoxylin, dehydrated, and mounted in nonaqueous mounting medium.

Immunization

Immunization of mice with CT (Sigma-Aldrich) was performed, as described previously (20). In brief, mice at 8 wk old were inoculated i.p. with 10 μg of CT in 0.2 ml of PBS emulsified with 0.2 ml of CFA. An identical dose of CT without CFA was given i.p. on day 14. On day 21, mice were deprived of food for 6 h and given 0.5 ml of an isotonic solution containing eight parts HBSS and two parts 7.5% sodium bicarbonate by gastric intubation to neutralize stomach acidity. After 30 min, mice were given 0.4 ml of PBS containing 10 μg of CT by gastric intubation, and also given 0.1 ml of PBS containing 10 μg of CT per rectum by intubation. Subsequently, mice were injected i.p. with 0.2 ml of PBS alone or PBS containing 100 μg of goat anti-mouse CCL25, goat anti-mouse CCL28, goat anti-human/mouse CXCL12, or normal goat IgG. On day 27, mice were sacrificed and lamina propria lymphocytes were isolated from small intestine and colon. Total and CT-specific IgA ASCs were enumerated with ELISPOT (see above). In parallel, serum samples and blood leukocytes were prepared. Serum CT-specific IgA and IgG titers were determined by ELISA and CT-specific IgA ASCs in blood leukocytes by ELISPOT.

ELISA

ELISA kits for mouse IgA and IgG were purchased from Bethyl Laboratories (Montgomery, TX). Serum CT-specific IgA and IgG titers were determined by ELISA using microtest plates precoated with CT. The endpoint titers were expressed as the reciprocals of the highest serum dilutions that gave an OD reading above the background level of 0.1.

Cell adhesion assay

Mouse MAdCAM-1-Fc and VCAM-1-Fc were purchased from R&D Systems. Control Fc fragment of human IgG1 was generated, as described previously (13). Cells were prepared from MLNs obtained from BALB/c mice and resuspended in adhesion medium (DMEM containing 0.5% BSA and 2 mM $MgCl_2$). Cells were preincubated without or with anti-mouse CD49d or control IgG (both from BD Pharmingen) for 30 min. After washing, cells were treated without or with CCL25 or CCL28 at 300 nM and immediately added in triplicate to 96-well microtest plates (High-binding; Corning Costar) precoated with 5 μg/ml Fc, MAdCAM-1-Fc, or VCAM-1-Fc at 1×10^6 cells/well. Plates were centrifuged for 15 s at 400 rpm and placed at 37°C for 5 min. Unbound cells were removed by gently washing with DMEM twice. Bound cells were released by incubation in ice-cold PBS containing 2 mM EDTA. IgA ASCs in original and adherent cell populations were then enumerated by ELISPOT.

Results

Differential expression of CCL25, CCL27, CCL28, and CXCL12 in various mouse organs and tissues

Because plasma cells are known to reside in a wide variety of organs, their immediate precursors, morphologically plasmablasts, and functionally ASCs are likely to migrate into various target tissues via locally produced chemokines. CCL25, CCL28, and CXCL12 are the chemokines known to play pivotal roles in plasma cell migration (6, 10, 13, 14, 17, 18). Even though the expression of these chemokines in various mouse tissues was separately examined by previous studies (11, 12, 15, 16, 21), we wished to directly compare their expression levels in various mouse organs and tissues. Therefore, we performed quantitative real-time PCR for CCL25, CCL28, and CXCL12. CCL27 was also examined because it shares CCR10 with CCL28 (22, 23). The results are shown in Fig. 1. CCL25 was selectively expressed in thymus, small intestine, and PP. CCL27, which is known to be selectively expressed in the skin (24, 25), was also expressed in thymus and weakly in brain and axillary LN (a draining LN of skin). CCL28 was strongly expressed in parotid gland, colon, and appendix. CCL28 was also moderately expressed in small intestine, PP, kidney, and brain. CCL28 was weakly expressed in thymus and lung and totally negative in liver, spleen, axillary LN, and MLN. Unexpectedly, CCL28 was also totally negative in mouse bone marrow, even though CCL28 was clearly shown to be expressed in human bone marrow (14). This may indicate a species difference in the role of CCL28 in bone marrow homing of plasma cells. Finally, CXCL12 was widely expressed in all the tissues examined, but its expression was very low in parotid gland and small intestine. Collectively, the highly differential expression of plasma cell-related chemokines in various organs and tissues supports their differential roles in plasma cell homing.

Surface expression of CCR9, CCR10, and CXCR4 on IgA ASCs

We next examined surface expression of CCR9, CCR10, and CXCR4, the receptors for CCL25, CCL28, and CXCL12, respectively, in IgA ASCs present in PP, MLN, small intestine, and colon (Fig. 2). CCR9, CCR10, and CXCR4 were detected by using chemokine-Fc chimera proteins (13). By gating on CD3-negative

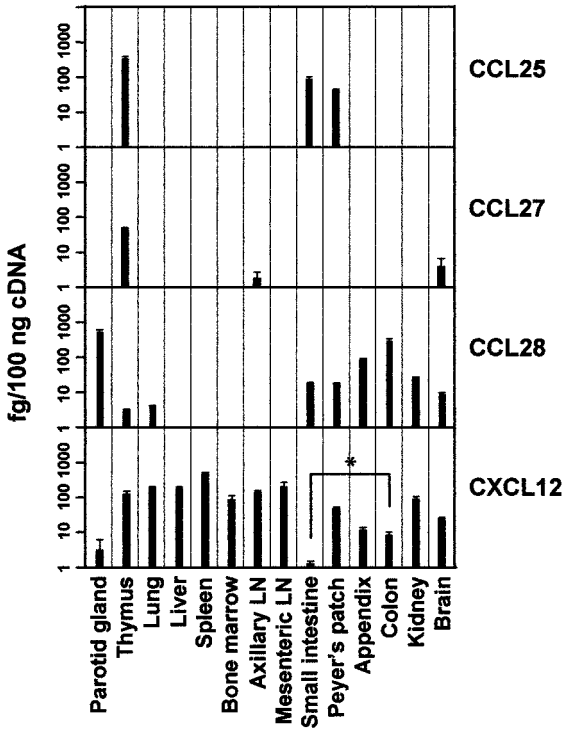

FIGURE 1. Expression of CCL25, CCL27, CCL28, and CXCL12 mRNA in various mouse organs and tissues. Total RNA samples were prepared from indicated mouse organs and tissues. Expression of CCL25, CCL27, CCL28, and CXCL12 mRNA was quantitated by real-time PCR. For details, see *Materials and Methods*. Data represent mean ± SD from three separate experiments. *, $p < 0.05$ (Student's *t* test).

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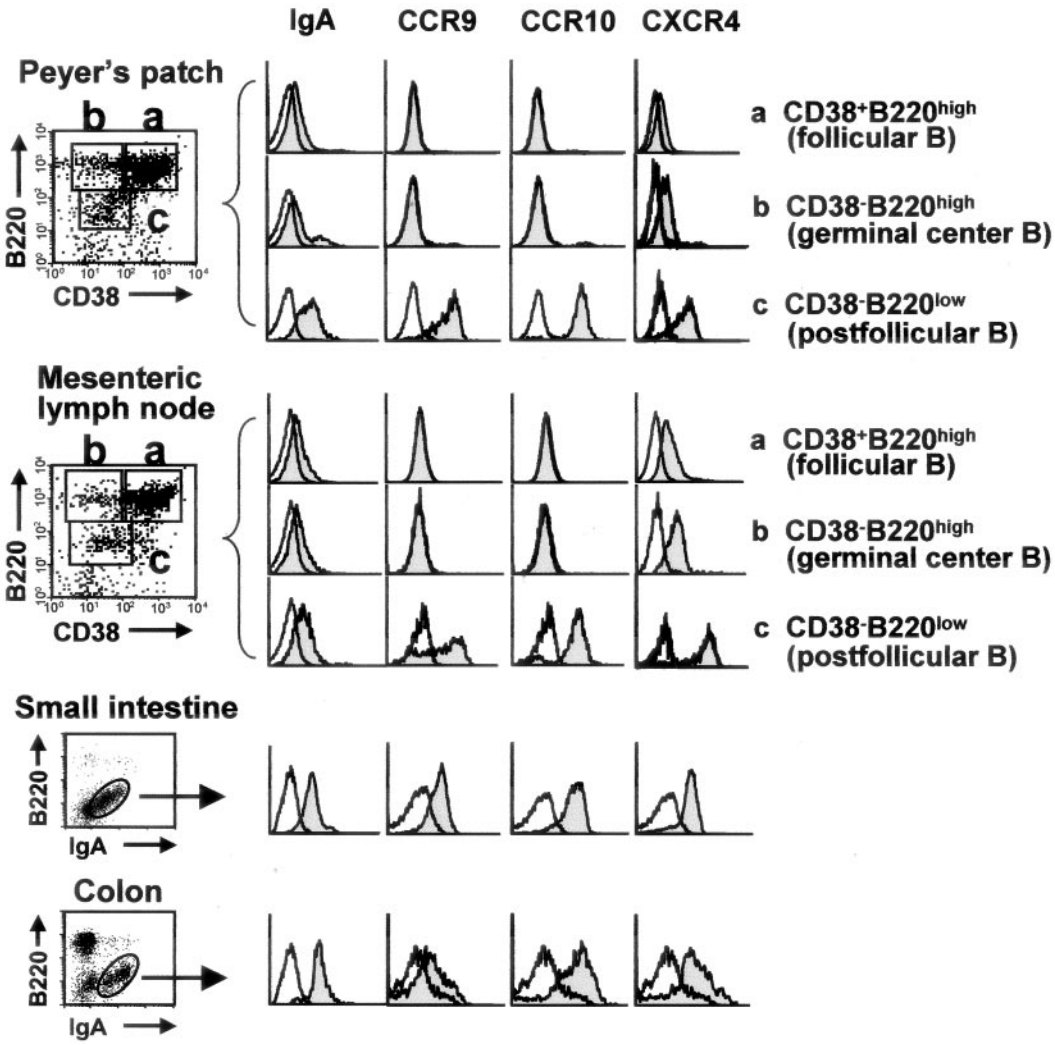

FIGURE 2. Flow cytometric analysis on surface expression of CCR9, CCR10, and CXCR4. Single cells prepared from PPs and MLNs were stained for CD3, B220, CD38, IgA, and CCR9, CCR10, or CXCR4. Single cells prepared from small intestine and colon were stained for CD3, B220, IgA, and CCR9, CCR10, or CXCR4. For surface staining of CCR9, CCR10, and CXCR4, cells were reacted with CCL25-Fc, CCL27-Fc, and CXCL12-Fc, respectively. For negative controls, cells were reacted with control Fc. The data were gated on CD3-negative populations. For details, see *Materials and Methods*. Representative results from at least three separate experiments are shown.

cells, $CD38^{+}B220^{high}$, $CD38^{-}B220^{high}$, and $CD38^{-}B220^{low}$ fractions in PP and MLN represented follicular B cells, germinal center B cells, and postfollicular B cells including plasma cells, respectively (26–28). Follicular B cells and germinal center B cells were essentially negative for CCR9 and CCR10 and weakly positive for CXCR4. In contrast, postfollicular B cells up-regulated surface IgA and were strongly positive for CCR9, CCR10, and CXCR4. Given a homogeneous expression pattern of surface IgA, CCR9, CCR10, and CXCR4 in postfollicular B cells of especially PP, the majority of these cells were likely to be IgA ASCs coexpressing CCR9, CCR10, and CXCR4. In MLN, however, a minor fraction of postfollicular B cells was negative for CCR9 or CCR10. This may indicate down-regulation of CCR9 and/or CCR10 in a fraction of IgA ASCs during migration into MLN. In small intestine, the fraction of $IgA^{+}B220^{low/-}$ cells represented the majority of lymphocytes in the lamina propria and were homogeneously positive for CCR9, CCR10, and CXCR4. Similarly, the fraction of $IgA^{+}B220^{low/-}$ cells in colon was again mostly positive for CCR9, CCR10, and CXCR4. These cells were likely to be IgA ASCs in the small intestine and colon (10). However, in comparison with $IgA^{+}B220^{low/-}$ cells in small intestine, those from colon expressed CCR9 at much lower levels. This may be in part due to selective recruitment of IgA ASCs expressing less CCR9 in the colon (17, 18).

Chemotactic responses of IgA ASCs to CCL25, CCL28, and CXCL12

We next examined chemotactic responses of IgA ASCs derived from PP, MLN, small intestine, and colon to CCL25, CCL28, and CXCL12 (Fig. 3). IgA ASCs were enumerated by ELISPOT. IgA ASCs in MLN vigorously responded to CCL28 and also to CCL25 and CXCL12. IgA ASCs in PP and small intestine responded to CCL25, CCL28, and CXCL12 at similar levels. Notably, however, IgA ASCs in colon responded to CCL28 and CXCL12, but not to CCL25. This was consistent with their low surface expression of CCR9 (Fig. 2). Taken together, in accordance with the surface expression of CCR9, CCR10, and CXCR4 (Fig. 2), IgA ASCs in these tissues were mostly capable of responding to CCL25, CCL28, and CXCL12. Furthermore, the highly efficient responses of IgA ASCs in MLN to CCL28 may indicate an enhanced signaling function of CCR10 in migrating IgA ASCs.

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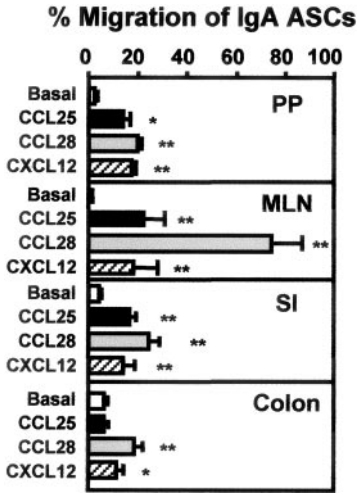

FIGURE 3. Chemotactic responses of IgA ASCs derived from various mouse tissues to CCL25, CCL28, and CXCL12. Single cells prepared from PPs, MLNs, small intestine (SI), and colon were examined for chemotactic responses to CCL25 at 300 nM, CCL28 at 300 nM, and CXCL12 at 50 nM. IgA-specific ELISPOT assay was performed to enumerate IgA ASCs in input and migrated cell populations. For details, see *Materials and Methods*. Data represent mean ± SD from three separate experiments. *, $p < 0.05$; **, $p < 0.01$ (Student's *t* test).

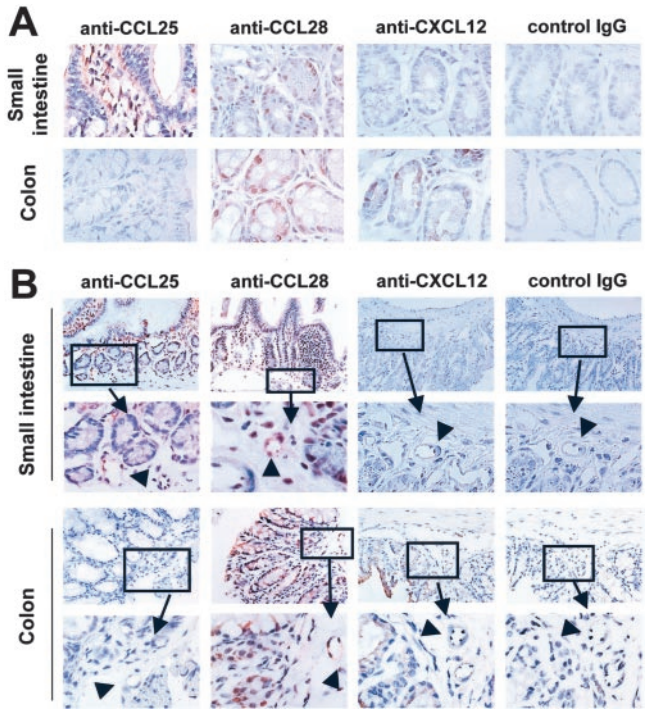

FIGURE 4. Immunohistochemistry of CCL25, CCL28, and CXCL12 in mouse small intestine and colon. Frozen sections of mouse small intestine and colon were fixed with periodate-lysine-4% paraformaldehyde and stained for CCL25, CCL28, and CXCL12 using goat anti-mouse CCL25, goat anti-mouse CCL28, and goat anti-human/mouse CXCL12, respectively. Negative controls were reacted with normal goat IgG. For details, see *Materials and Methods*. Representative results from at least three separate experiments are shown. *A*, Immunoreactivity of epithelial cells. Original magnification: ×400. *B*, Immunoreactivity of endothelial cells. Original magnifications: *upper panels*, ×400; *lower panels*, ×1000. Arrowheads indicate endothelial cells of microvessels.

Immunohistochemical staining of CCL25, CCL28, and CXCL12

To determine cells producing and/or presenting CCL25, CCL28, and CXCL12 in the small intestine and colon, we next performed immunohistochemical staining. As shown in Fig. 4*A*, the epithelial cells of small intestine were positive for CCL25 and CCL28. In contrast, the epithelial cells of colon were positive for CCL28 and CXCL12. Furthermore, as shown in Fig. 4*B*, the venular endothelial cells in small intestine were positive for CCL25 and CCL28, while those in colon were positive for CCL28. These results were highly consistent with the differential expression of CCL25, CCL28, and CXCL12 mRNA in the small intestine and colon (Fig. 1). Furthermore, the endothelial staining of CCL25 in small intestine and CCL28 in both small intestine and colon may indicate their direct roles in extravasation of IgA ASCs by arresting IgA ASCs on the endothelial cell surface and promoting their transendothelial migration into intestinal lamina propria. In contrast, CXCL12 may not be directly involved in extravasation of IgA ASCs into intestinal lamina propria.

Effects of anti-CCL25, anti-CCL28, and anti-CXCL12 on intestinal homing of IgA ASCs

To explore the roles of CCL25, CCL28, and CXCL12 in intestinal homing of plasma cells, we next examined effects of in vivo neutralization of CCL25, CCL28, and CXCL12 on homing of Ag-specific IgA ASCs into small intestine and colon in mice immunized with CT. CT-specific IgA ASCs and total IgA ASCs were enumerated by ELISPOT. As shown in Fig. 5*A*, both anti-CCL25 and anti-CCL28 significantly reduced CT-specific IgA ASCs among total IgA ASCs in small intestine. No such reduction was seen with anti-CXCL12 or control IgG. This supported that CCL25 and CCL28, but not CXCL12, were involved in the extravasation of newly generated CT-specific IgA ASCs into small intestine. Rather unexpectedly, however, the combined treatment with anti-CCL25 and anti-CCL28 did not further reduce CT-specific IgA ASCs in small intestine. In the case of colon, only anti-CCL28 significantly reduced CT-specific IgA ASCs among total IgA ASCs. No such reduction was seen with anti-CCL25, anti-CXCL12, or control IgG. No further reduction was seen by the combined treatment with anti-CCL25 and anti-CCL28 either. If these results indicated that intestinal extravasation of newly generated CT-specific IgA ASCs was blocked in mice treated with anti-CCL25 or anti-CCL28, there would be reciprocal increases in CT-specific IgA ASCs and titers in the blood of these mice. This was indeed the case. As shown in Fig. 5*B*, mice treated with anti-CCL25 or anti-CCL28, but not those treated with anti-CXCL12, showed significant increases in CT-specific IgA ASCs in the blood. We were unable to compare CT-specific IgG ASCs because of their very low numbers in the blood of these mice (data not shown). Furthermore, as shown in Fig. 5*C*, serum CT-specific IgA titers, but not CT-specific IgG titers, were significantly increased in mice treated with anti-CCL25 or anti-CCL28, but not in mice treated with anti-CXCL12. In both parameters, no further increases were seen by the combined treatment with anti-CCL25 and anti-CCL28. Rather unexpectedly, however, there were no significant differences in increase of CT-specific IgA ASCs and titers between mice treated with anti-CCL25 and those treated with anti-CCL28, even though the latter would be expected to block extravasation of CT-specific IgA ASCs in both small intestine and colon. This was probably due in part to the vast numerical dominance (~10 times) of total CT-specific IgA ASCs homing into small intestine over those homing into colon (data not shown). Collectively, these data supported that CCL25 and CCL28 play a direct role in the extravasation of IgA ASCs in small intestine (CCL25 and CCL28) and colon (CCL28).

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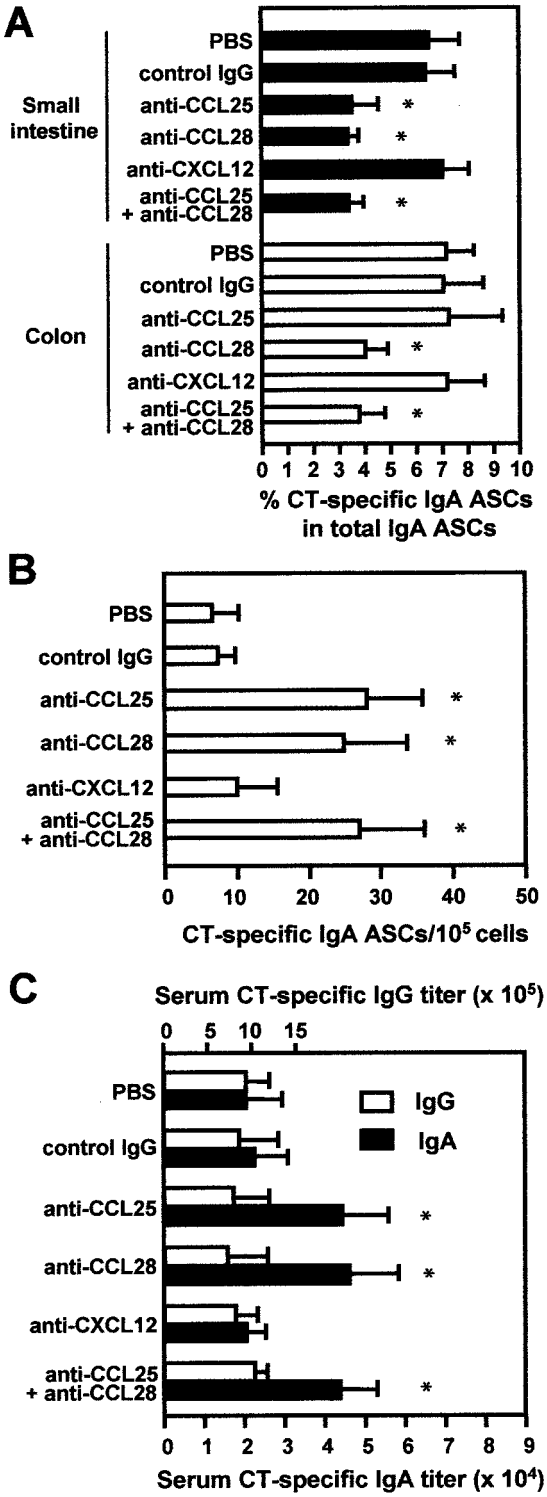

FIGURE 5. Effects of blocking CCL25, CCL28, and CXCL12 on intestinal homing of Ag-specific IgA ASCs. BALB/c mice primed with CT were immunized orally and per rectum with CT and also injected i.p. with PBS, normal goat IgG, goat anti-mouse CCL25, goat anti-mouse CCL28, or goat anti-human/mouse CXCL12, as indicated. After 6 days, lymphocytes were prepared from lamina propria of small intestine and colon. Serum samples and blood leukocytes were also prepared. Each group consisted of at least five mice. Statistical analysis was done with Student's *t* test. *A*, Intestinal homing of CT-specific IgA ASCs. Cells were prepared from lamina propria of small intestine and colon. CT-specific IgA ASCs and total IgA ASCs were enumerated by ELISPOT. For details, see *Materials and Methods*. Data represent mean ± SD from four separate experiments. *, $p < 0.05$. *B*, CT-specific IgA ASCs in peripheral blood. CT-specific IgA ASCs in peripheral blood leukocytes were enumerated by ELISPOT. For details, see *Materials and Methods*. Data represent mean ± SD from three separate experiments. *, $p < 0.05$. *C*, Serum CT-specific IgG and IgA titers. CT-specific IgG titers (□) and IgA titers (■) were determined with ELISA. For details, see *Materials and Methods*. Data represent mean ± SD from four separate experiments. *, $p < 0.05$.

CCL25 and CCL28 induce α_4 integrin-dependent adhesion of IgA ASCs to MAdCAM-1 and VCAM-1

Plasma cells homing to intestine are known to express the mucosal homing receptor $\alpha_4\beta_7$ integrin (29), whereas its ligand MAdCAM-1 is predominantly expressed by postcapillary venules in intestinal mucosal tissues (30). In mice lacking β_7, plasma cell numbers in the intestinal lamina propria were greatly reduced (31). Furthermore, VCAM-1, which is another ligand of $\alpha_4\beta_7$ integrin, is expressed on the endothelium of nonintestinal mucosal tissues that also express CCL28 (5). Given the presumed roles of CCL25 and CCL28 in extravasation of IgA ASCs in intestinal tissues, we next asked whether these chemokines were capable of inducing firm adhesion of IgA ASCs to MAdCAM-1 and VCAM-1 in vitro. We tested IgA ASCs in MLN as responding cells because MLN contained IgA ASCs vigorously responding to CCL25 and CCL28 in chemotaxis assays (Fig. 3). Cell adhesion assays were performed under static conditions, and total and bound IgA ASCs were enumerated with IgA-specific ELISPOT. As shown in Fig. 6, both CCL25 and CCL28 significantly enhanced adhesion of IgA ASCs to MAdCAM-1 and VCAM-1, with CCL28 being more efficient than CCL25. Furthermore, anti-α_4 (CD49d), but not control IgG, effectively blocked CCL25- and CCL28-induced adhesion of IgA ASCs to MAdCAM-1 and VCAM-1, confirming activation of $\alpha_4\beta_7$ integrin expressed on IgA ASCs by CCL25 and CCL28.

Discussion

IgA plasma cells play pivotal roles in mucosal immunity against invading microorganisms. Although IgA plasma cells generated in small intestine and colon preferentially home into small intestine and colon, respectively (32), those generated by immunization via nasal cavity preferentially home into nasopharyngeal and respiratory mucosal tissues, but not into intestinal tissues (33, 34). Such differential migratory properties of IgA plasma cells in accordance with their tissue origin are likely to be explained in part by their expression of a different set of chemokine receptors (5). Recently, Butcher and his coworkers (10, 17, 18) have elegantly demonstrated the differential roles of CCL25 and CCL28 in the mucosal immunity. Although CCL25 is selectively expressed in the small intestine, its receptor CCR9 is expressed by almost all T cells in the small intestine and a fraction of IgA ASCs. In contrast, CCL28 is widely expressed in intestinal and nonintestinal mucosal tissues, and its receptor CCR10 is expressed by almost all IgA ASCs. Notably, however, T cells homing to intestinal tissues hardly express CCR10. In fact, T cells expressing CCR10 are CLA^+ memory/effector T cells, which preferentially migrate to skin where epidermal keratinocytes produce CCL27, another ligand of CCR10 (18, 24, 35). Thus, the CCL25/CCR9 axis may play a specific role in small intestine by attracting memory/effector T cells and a fraction of IgA ASCs directed to small intestine, whereas the CCL28/CCR10 axis may play a more unifying role in the common mucosal immune system by recruiting IgA ASCs to a wide variety of mucosal tissues (10, 17, 18).

In the present work, we have further explored the roles of chemokines in intestinal homing of IgA plasma cells. First, we have demonstrated highly differential expression of CCL25, CCL27, CCL28, and CXCL12 mRNA in various mouse tissues, including

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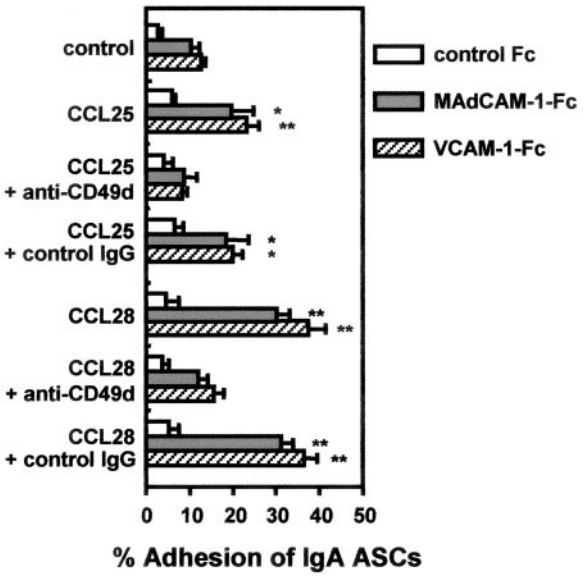

FIGURE 6. Induction of α_4 integrin-dependent adhesion of IgA ASCs to MAdCAM-1 and VCAM-1 by CCL25 and CCL28. Single cells were prepared from MLNs. Cells were treated without or with CCL25 or CCL28 at 300 nM and immediately added to microtest plates coated with MAdCAM-1-Fc, VCAM-1-Fc, or control Fc. After incubation at 37°C for 5 min, nonadherent cells were gently washed away. IgA ASCs in input and adherent cell populations were enumerated by IgA-specific ELISPOT. For details, see *Materials and Methods*. Data represent mean ± SD from three separate experiments. *, $p < 0.05$; **, $p < 0.01$ (Student's *t* test).

small intestine and colon (Fig. 1). Second, we have demonstrated that IgA ASCs in PP, MLN, small intestine, and colon mostly coexpress CCR9, CCR10, and CXCR4 on their cell surface (Fig. 2) and are mostly capable of responding to CCL25, CCL28, and CXCL12 in chemotaxis assays (Fig. 3). Thus, IgA ASCs in these tissues can be guided by any of these chemokines alone or in combination. However, given the highly differential expression of CCL25, CCL28, and CXCL12 in tissues such as PP, MLN, small intestine, and colon (Fig. 1), it is likely that IgA ASCs entering into MLN are mainly guided by CXCL12, whereas those homing into small intestine are guided by CCL25 and CCL28, and yet those homing into colon are guided by CCL28 and CXCL12. Consistently, IgA ASCs in the colon expressed CCR9 at low levels (Fig. 2) and poorly responded to CCL25 in chemotaxis assays (Fig. 3) (10, 17, 18). Thus, a subset of IgA ASCs, possibly generated in the colon, may express CCR9 only at low levels. Third, we have immunologically demonstrated the presence of CCL25 in the epithelial cells of small intestine, CCL28 in the epithelial cells of both small intestine and colon, and CXCL12 in the epithelial cells of colon (Fig. 4*A*). CCL25 was reported to be localized in the crypts of Lieberkühn in human small intestine (12). In mouse small intestine, however, we detected CCL25 more widely in villus epithelial cells and within the villi. Thus, in mice, CCL25 may be produced by most villus epithelial cells and secreted into subepithelial tissue spaces. As noted previously in human small intestine (12), we have also demonstrated the presence of CCL25 on the venular endothelium of mouse small intestine. Furthermore, we have demonstrated for the first time the presence of CCL28 on the venular endothelium of both small intestine and colon in mice (Fig. 4*B*).

The above mentioned results strongly suggested that CCL25 plays a direct role in extravasation of IgA ASCs in small intestine, while CCL28 does so in both small intestine and colon. We have indeed demonstrated that goat anti-CCL25 dramatically reduced homing of Ag-specific IgA ASCs into small intestine, whereas goat anti-CCL28 reduced homing of Ag-specific IgA ASCs into both small intestine and colon (Fig. 5*A*). In contrast, no such reduction in homing of Ag-specific IgA ASCs was seen in mice treated with goat anti-CXCL12. Reciprocally, there were selective increases in Ag-specific IgA ASCs and Ag-specific IgA titers in the blood of mice treated with anti-CCL25 or anti-CCL28 (Fig. 5, *B* and *C*). Furthermore, we confirmed that both CCL25 and CCL28 were capable of inducing adhesion of freshly isolated IgA ASCs to MAdCAM-1 and VCAM-1 via activation of $\alpha_4\beta_7$ integrin (Fig. 6). These results strongly support the notion that the extravasation of IgA ASCs into small intestine is mediated by both CCL25 and CCL28, whereas the extravasation of IgA ASCs into colon by CCL28. Collectively, CCL25 and CCL28 are likely to play essential roles in intestinal homing of IgA ASCs primarily by promoting their extravasation from small venules into intestinal lamina propria. After that, CCL25 and CCL28 produced by epithelial cells of small intestine may further navigate IgA ASCs within lamina propria of small intestine, whereas CCL28 and CXCL12 produced by epithelial cells of colon may guide IgA ASCs within lamina propria of colon (Fig. 4*A*). Because of a relatively low neutralizing potency of anti-CXCL12 in comparison with those of anti-CCL25 and anti-CCL28 (see *Materials and Methods*), however, we could not formally exclude a role of CXCL12 in extravasation of IgA ASCs, especially in the colon. Furthermore, we could not formally exclude some nonspecific effects of injected goat Abs on the homeostatic regulation of IgA ASCs via Fc receptors and/or complement receptors. Thus, generation of mice with targeted disruption of CCL25 and CCL28 genes would be necessary to further define their roles in the intestinal homing of IgA ASCs.

By using 24-h in vivo homing assays, Reiss et al. (36) demonstrated that either CCL27 (a CCR10 ligand selectively expressed by epidermal keratinocytes) or CCR4 (probably via its ligand CCL17 selectively expressed by cutaneous microvascular endothelial cells) was sufficient for recruitment of skin-homing memory T cells into inflamed skin. Thus, CCL27 and CCR4 apparently play a redundant role in recruitment of activated memory T cells in inflamed skin. In this study, in contrast, blocking of either CCL25 or CCL28 similarly reduced homing of Ag-specific IgA ASCs in small intestine. Thus, CCL25 and CCL28 appear to play a nonredundant role in recruitment of IgA ASCs into small intestine. This may be in part due to their relatively low levels of expression in small intestine (Fig. 1).

Recently, by using CCR9-deficient mice, Pabst et al. (37) have also demonstrated that CCR9 contributes to the localization of plasma cells to the small intestine. In these mutant mice, IgA plasma cells in lamina propria were reduced by ~50% in small intestine, but not in colon. Thus, their observations were highly consistent with our findings and also support that CCL28/CCR10 signals do not completely compensate for CCL25/CCR9 signals in homing of IgA ASCs in small intestine. In CCR9-deficient mice, however, serum OVA-specific IgA titers were barely elevated even after repeated oral immunization with OVA (37). This suggests that not only homing of IgA plasma cells, but also T cell-dependent immune responses in small intestine were severely impaired in CCR9-deficient mice. In this study, however, we demonstrated increases in CT-specific IgA titers and IgA ASCs in the blood of mice treated with anti-CCL25 or anti-CCL28 (Fig. 5, *B* and *C*). This suggests that immune responses to orally administered CT were mostly intact, but homing of CT-specific IgA ASCs into small intestine and/or colon was acutely impaired in these mice. Unexpectedly, the combined treatment with anti-CCL25 and anti-CCL28 did not further reduce homing of Ag-specific IgA ASCs in small intestine (Fig. 5*A*). Homing of IgA ASCs was also only partially blocked by anti-CCL28 in colon (Fig. 5*A*). Even though

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we could not exclude incomplete neutralization of CCL25 or CCL28 in mice treated with anti-CCL25 or anti-CCL28, these results might indicate the presence of long-lived CT-specific IgA ASCs in these mice that were systematically preimmunized with CT. Another possibility may be that some IgA ASCs are still capable of entering into intestinal tissues by responding to other chemotactic factors. These possibilities remain to be seen.

Finally, we observed some mononuclear cells strongly positive for CCL28 in PPs and lamina propria of small intestine (Fig. 4*B*). The identity of these cells remains to be seen. It also remains to be seen whether CCL25 and/or CCL28 are directly produced by endothelial cells in the small intestine and colon or are produced by epithelial cells, transported to endothelial cells in the lamina propria, and presented on their cell surface. We have also demonstrated a highly restricted expression of CCR9 and CCR10 in postfollicular B cells in PP and MLN (Fig. 2). Thus, CCR9 and CCR10 are expressed only at the terminal stages of B cell differentiation. The molecular mechanisms of restricted expression of CCR9 and CCR10 in the plasma cell stage remain to be seen. It also remains to be seen how and to what extent CCR9 and CCR10 are restricted to IgA plasma cells than those producing other classes of Ig. Furthermore, the signaling functions of CCR9, CCR10, and CXCR4 may also be dynamically regulated along the process of plasma cell differentiation and homing (Fig. 3) (8). These possibilities remain to be addressed.

Acknowledgment

We thank Namie Sakiyama for excellent technical assistance.

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